

root node endpoint, which SHALL support the Secondary Network Interface device type.

A Node supporting secondary network interfaces SHALL support the concurrent connection commissioning flow (see [Figure 45, “Commissioning flow diagram”](#)). Therefore, if the primary network configuration fails, the Commissioner can proceed to configure the secondary network interfaces via the established commissioning channel.

11.9.8.2. Order of connectivity during connection establishment

When at least one Network Commissioning cluster server instance (hereafter, "Network Commissioning cluster" for short), the following behavior SHOULD take place for each interface associated with a Network Commissioning cluster, in increasing order of associated endpoint number:

1. If the Network Commissioning cluster's InterfaceEnabled attribute is set to false, skip the processing the interface altogether.
2. Set all configurations of the [Networks attribute](#) entry's Connected field to false
3. Iterate through all configurations in the [Networks attribute](#)
 - a. If there was a "last known good" network configuration, that is, the one which was both last successfully connected during prior boot and over which at least one secure channel exchange message was received, it MAY be used as the first attempt. Otherwise, iterate through all configurations in the precedence order of the list, starting at index 0.
4. Attempt to connect to the technology, using the current iteration's network configuration
5. On success, set the Connected state of the list entry to true, and stop attempting further connection. Otherwise, on failure, move to the next configuration.

11.9.8.3. Connectivity management during commissioning or administration

When a network interface is configured during commissioning or reconfigured during ongoing administration, behavior is different than for the startup case described previously, since there are tentative attempts being made to make a Node reachable on an operational network.

Network configuration can be seen as:

- A list of existing configurations, reflected by the [Networks attribute](#).
 - The list SHALL be managed by the AddOrUpdateWiFiNetwork, AddOrUpdateThreadNetwork, RemoveNetwork and ReorderNetwork commands.
 - The list SHALL be tentative until committed by successful invocation of the [Commissioning-Complete](#) command, or reverted to prior configuration by the expiry of the Fail-Safe timer (see [Section 11.10.7.2, “ArmFailSafe”](#)).
- A current candidate configuration, the subject of the most recent ConnectNetwork command.
 - There SHALL NOT be new connections to any network during the Fail-Safe timer period unless attempted by invocation of the ConnectNetwork command.

Failures of connection during the Fail-Safe timer window SHALL cause the Node to follow the steps in [Section 5.5.1, “Commissioning Flows Error Handling”](#) after recording the cause of the failure in the [LastNetworkingStatus](#), [LastNetworkID](#) and [LastConnectErrorValue](#) attributes.